

Response to Reviewer Comments

Manuscript #18579

“Byzantine-Robust Federated Learning with Adaptive Aggregation and Blockchain:
Empirical Validation of ATMA and Resolution of the Transparency Paradox”

April 13, 2026

We sincerely thank the editor and reviewers for their constructive feedback. We have carefully addressed every comment below. All changes in the revised manuscript are marked with **blue text** where feasible. Page and section numbers refer to the revised manuscript that follows this response.

Reviewer A

Comment A1: *“Attack types are limited to label-flip and random noise; sophisticated attacks (ALIE [18], backdoor [20]) require future study.” rephrase this as the citations look strange now*

Answer: We agree that the inline citation format was unclear. We have rephrased the sentence in Section 7 (Limitations) as follows:

“Attack types are limited to random noise and label-flip attacks. More sophisticated attack strategies, such as the ‘A Little Is Enough’ (ALIE) approach proposed by Baruch et al. [10] and backdoor attacks studied by Sun et al. [11], were not fully evaluated in this work and require future study.”

The citation numbers now clearly refer to the named authors and their specific attack methods, eliminating the ambiguity.

Comment A2: *Figures 1-2 not readable.*

Answer: We have fully regenerated both figures:

- **Figure 1 (Convergence Comparison):** Regenerated at 300 DPI with distinct markers (square, triangle, diamond, circle), distinct colors, thicker lines (2.0–2.5 pt), and larger axis labels (13 pt bold). The figure now clearly shows CIFAR-10 convergence curves for all four algorithms (TrimmedMean, ATMA, Krum, FedAvg) over 160 rounds, with final accuracy values in the legend.
- **Figure 2 (ROC Curve):** Regenerated at 300 DPI with thicker curve lines (3.0 pt for blockchain, 2.5 pt for centralized), annotated optimal operating point (FPR=0.00, TPR=0.81), 13 pt bold axis labels, and clear legend showing AUC values.

Both figures are now publication-quality at 300 DPI with clearly distinguishable elements. The previous version suffered from low resolution and thin lines that became illegible in the Word document conversion.

Comment A3: *Conclusion (and article) too short I think.*

Answer: We have substantially expanded the Conclusion section from one page to approximately three pages. The expanded conclusion now includes:

- **Section 7.1 (Summary of Key Findings):** Four clearly numbered findings with specific accuracy values and statistical tests.
- **Section 7.2 (Comparative Analysis with Prior Work):** A new Table XII comparing our contributions against Blanchard et al. [3], Yin et al. [4], Shayan et al. [27], Kalibbala et al. [46], and Cao et al. [35] across six dimensions (robust aggregation, blockchain audit, adaptive defense, non-IID validation, transparency paradox, multi-seed CI).
- **Section 7.3 (Practical Contributions and Deployment Guidelines):** Concrete recommendations for algorithm selection, cost management, scalability, and security posture.
- **Section 7.4 (Limitations):** Honest enumeration of four specific limitations.
- **Section 7.5 (Future Work):** Six concrete future research directions.

Comment A4: *Expand it to include comparative analysis, contribution discussion.*

Answer: Addressed in Comment A3 above. Table XII in the revised conclusion provides a structured comparative analysis against five key prior studies, and the “Practical Contributions” subsection explicitly discusses what each contribution means for practitioners deploying Byzantine-robust federated learning.

Reviewer B

Comment B1: *It is not clear what EXACTLY the authors did and how (step by step).*

Answer: We acknowledge this was a significant weakness in the original manuscript. We have added a new subsection **Section 4.6 “Step-by-Step Experimental Procedure”** that describes the exact procedure we executed, with no generic terms:

1. **Data Loading:** Downloaded MNIST via `torchvision.datasets.MNIST` and CIFAR-10 via `torchvision.datasets.CIFAR10`; normalized with dataset-specific μ and σ values.
2. **Non-IID Partitioning:** Applied `Dirichlet( $\alpha = 0.5$ )` allocation using `numpy.random.dirichlet` to create heterogeneous per-client data splits.
3. **Model Initialization:** SimpleMNISTNet for MNIST (2 Conv + 2 FC); ResNet-18 for CIFAR-10. Same seed for identical initialization.
4. **Byzantine Client Selection:** Randomly selected 4 of 20 clients as Byzantine, fixed per seed.
5. **Blockchain Contract Deployment:** Deployed `FederatedLearningAggregator` Solidity contract to Ganache using `solcx`.
6. **Training Loop** (per round): client selection $\rightarrow$ local SGD training $\rightarrow$ Byzantine attack injection (label-flip + scale $\lambda = -5.0$) $\rightarrow$ blockchain submission (SHA-256 hash) $\rightarrow$ server-side aggregation $\rightarrow$ Byzantine detection logging (on-chain) $\rightarrow$ global model update $\rightarrow$ test evaluation.
7. **Result Serialization:** All metrics saved to JSON.

8. **Multi-Seed Replication:** Seeds 42, 43, 44 for confidence intervals.

The detailed 8-step procedure in the revised manuscript specifies every operation, parameter value, and code-level function, so that another researcher can reproduce our work exactly.

Comment B2: *There is no point in throwing a lot of terms without being specific on what you did exactly and how. Not generic information. SPECIFICALLY in this work.*

Answer: We completely agree. Beyond the new Step-by-Step procedure (Section 4.6), we have reviewed the entire paper to ensure specificity. Key improvements:

- **Every aggregation algorithm** is described with its exact formula and our configuration (e.g., TrimmedMean: $\beta = 0.2$, meaning top/bottom 20% trimmed per coordinate; ATMA: adaptation rate $\alpha = 0.05$, bounds $[\tau_{\min}, \tau_{\max}] = [0.05, 0.30]$).
- **Every experiment** references the **specific result file** in our public repository (Table I lists file names: `h2_valid_rerun.json`, `real_fl_FIXED.json`, etc.).
- **Specific data paths:** We state that MNIST is loaded “via `torchvision.datasets.MNIST`” (not just “MNIST dataset”), Dirichlet partitioning is done “using `numpy.random.dirichlet`” with $\alpha = 0.5$.
- **Blockchain operations:** Contract deployment gas (1,724,238 gas), per-submission gas ($\approx 75,000$ gas), and each on-chain function call (`submitUpdate`, `markByzantine`, `completeRound`) are explicitly described.
- **All source code** is available at <https://github.com/vokasitibrawijaya/byzantine-robust-fl-blockchain> for full reproducibility.

— REVISED MANUSCRIPT FOLLOWS —

See separate file: 1kolomByzantineRobustFederatedLearning_
withAdaptiveAggregationandBlockchainFIXED.tex